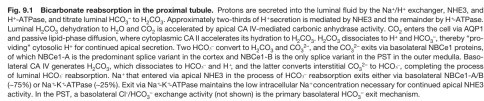
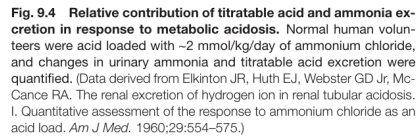


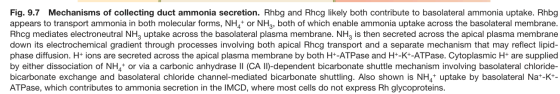
09. Renal Acidification Mechanisms · assets 1-7 / 7



Fig_9_1



Fig_9_4



Fig_9_7

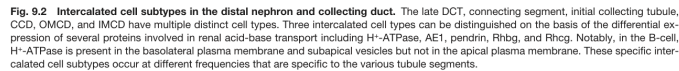


Fig. 9.2 **Intercalated cell subtypes in the distal nephron and collecting duct.** The late DCT, connecting segment, initial collecting tubule, CCD, OMCd, and IMCD have multiple distinct cell types. Three intercalated cell types can be distinguished on the basis of the differential expression of several proteins involved in renal acid-base transport including H⁺-ATPase, AE1, pendrin, Rhbg, and Rhcg. Notably, in the B-cell, H⁺-ATPase is present in the basolateral plasma membrane and subapical vesicles but not in the apical plasma membrane. These specific intercalated cell subtypes occur at different frequencies that are specific to the various tubule segments.



Fig_9_3

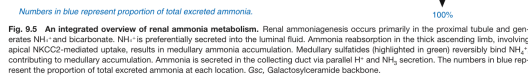


Fig. 9.5 An integrated overview of renal ammonia metabolism. Renal ammoniogenesis occurs primarily in the proximal tubule and generates NH_4^+ and bicarbonate. NH_4^+ is preferentially secreted into the luminal fluid. Ammonia reabsorption in the thick ascending limb, involving apical NKCC2 -mediated uptake, results in medullary ammonia accumulation. Medullary sulfates (highlighted in green) reversibly bind NH_4^+ , contributing to medullary accumulation. Ammonia is secreted in the collecting duct via parallel H^+ and NH_3 secretion. The numbers in blue represent the proportion of total excreted ammonia at each location. Gsc, Galactose/ceramide backbone.

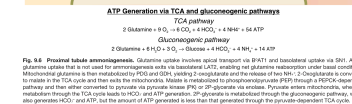


Fig. 9b. Proximal tubule ammonioexcretion. Glutamine uptake involves active transport via B⁰AT1 and basolateral uptake via SNT. A glutamine uptake that is not used for ammonioexcretion is used via basolateral LAT2, entering renal glutamine reabsorption under basal conditions. Glutamine glutamate is then metabolized by GDC and GDH, yielding 2 moles of NH₄⁺ and 2 moles of NH₃. One glutamate is converted to malate in the TCA-cycle and then exits the mitochondria. Malate is metabolized to phosphoenolpyruvate (PEP) through a PEPCK-dependent pathway and then either converted to pyruvate via pyruvate kinase (PK) or 2-P-glycinate via enolase. Pyruvate enters mitochondria, where metabolism through the TCA cycle leads to HCO₃⁻ and ATP generation. 2-P-glycinate is metabolized through the gluconeogenic pathway, also generates HCO₃⁻ and ATP, but the amount of ATP generated is less than that generated through the pyruvate dependent TCA cycle.

Fig_9_6